

Digital Solutions for Sustainable Agriculture and Community Development with Wassan Foundation

Shayantan Bose

Dr. C. Arun Kumar
Assistant Professor
School of Sciences,
Division of
Mathematics

Contents

- Problem Statement
- Details of NGO visited (Name, Address, DARPAN ID)
- Thematic Area and its mapping to SDG
- Proposed Engineering Solution
- Non-profit alignment
- Proposed Milestones and Timelines
- Expected Deliverables (Prototypes, Reports, Designs, Viable Solutions)
- Plans to Assess the Impact
- Proposed Budget
- Conclusion

Problem Statement

Problem Description:

Rural agricultural communities and NGOs like Wassan Foundation face significant challenges in:

- Manual bill processing and financial record keeping leading to inefficient resource management
- Limited digital presence to promote sustainable agricultural practices and natural farming
- Lack of accessible platforms to educate communities about chemical-free living and environmental consciousness

Problem Definition Through Field Work:

- Field visits to Wassan Foundation revealed manual processes for expense tracking and bill management
- Surveys indicated need for digital platforms to reach wider audiences interested in sustainable living
- User stories from NGO staff highlighted time-consuming administrative tasks

Target Audience/Users:

- Wassan Foundation staff and administrators
- Rural farmers interested in natural farming practices
- Urban communities seeking sustainable living solutions
- Researchers and organizations in agricultural sustainability

Number of Users: Estimated 500+ direct users (NGO staff, partner organizations) and 10,000+ indirect beneficiaries through improved

Details of the NGO

Name and Address: Wassan Foundation (Watershed Support Services and Activities Network)

Website: <https://www.wassan.org/>

Nature of Activities:

- Anchors the Revitalising Rainfed Agriculture Network (RRA Network)
- Leads the National Coalition on Natural Farming (NCNF) Secretariat
- Promotes sustainable agricultural practices and natural farming
- Community development in rural areas
- Environmental conservation and watershed management

DARPAN ID: AP/2016/0107413

Communities Served: Rural farming communities across India, particularly focusing on rainfed agriculture areas

Key Contact Person: MV MV Ramachandrudu

WATERSHED SUPPORT SERVICES AND ACTIVITIES NETWORK	
NPO Details	
DARPAN ID AP/2016/0107413	DARPAN Registration Date --
Registration Details	
Registered With Sub-Registrar	Type of NPO Trust
Registration No 143/118/IV/2021	Act name Public Charitable Trust
City of Registration Others	State of Registration ALL
Date of Registration (Society / Trust / Entity) 20-12-1999	

Thematic Area

Sustainable Development Goals Addressed:

1. **SDG 1 - No Poverty:** Improving efficiency of NGO operations leads to better resource allocation for poverty alleviation
2. **SDG 2 - Zero Hunger:** Supporting natural farming practices through digital platforms enhances food security
3. **SDG 3 - Good Health:** Promoting chemical-free living and sustainable practices improves community health
4. **SDG 8 - Decent Work:** Creating digital solutions that improve organizational efficiency
5. **SDG 9 - Industry Innovation:** Implementing innovative digital solutions for traditional NGO operations
6. **SDG 12 - Responsible Consumption:** Promoting sustainable living through digital awareness platforms
7. **SDG 15 - Life on Land:** Supporting natural farming and environmental conservation efforts

Proposed Engineering Solution

Develop three integrated digital solutions to enhance Wassan Foundation's impact and operational efficiency.

Technical Modules:

Module 1: Bill Management Web Application

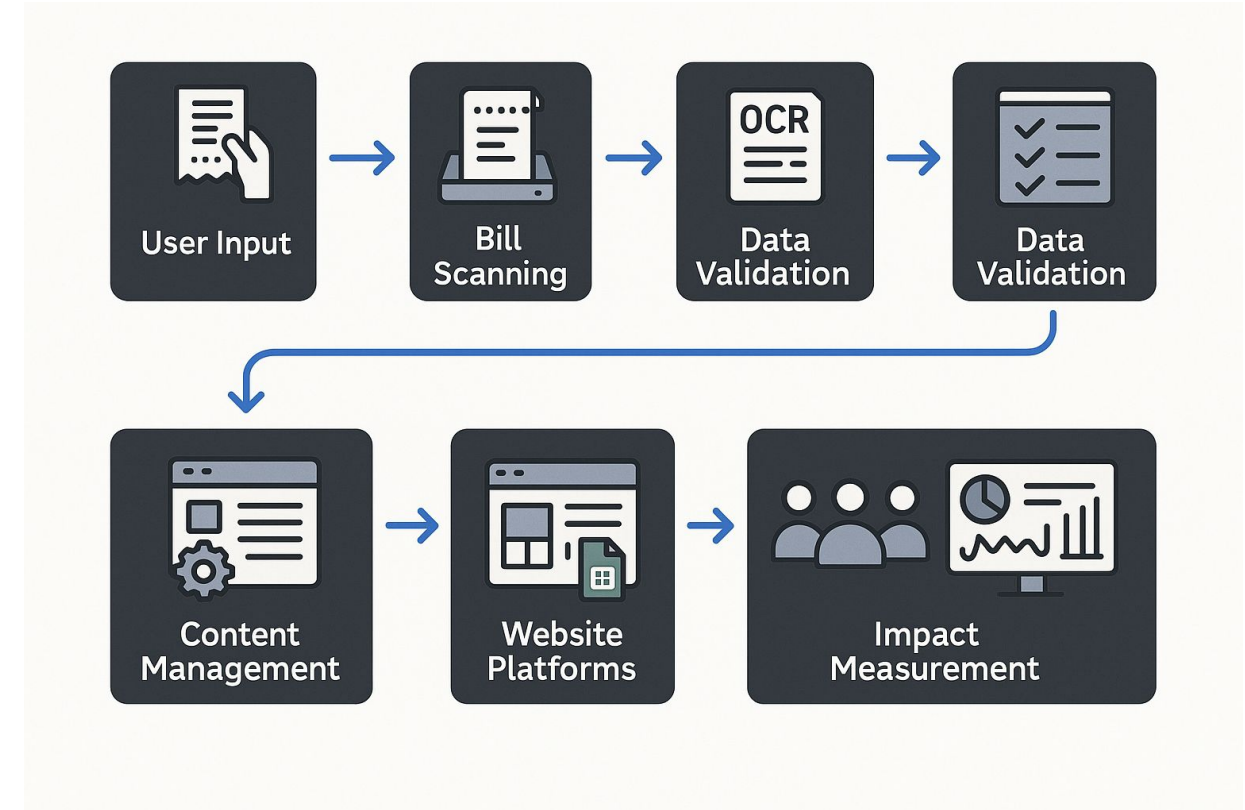
- OCR-based bill scanning system
- Google Sheets API integration
- Collaborative expense tracking dashboard

Module 2: Sonta Dabba Manava Lokkam Website

- Content management system for sustainable living content
- Community engagement features
- Resource library for chemical-free living

Module 3: Wassan Foundation Website Redesign

- Modern, responsive design
- Enhanced user experience for stakeholders
- Mobile-optimized interface



Non-Profit Alignment

NGO Needs:

- Streamlined financial management and expense tracking
- Enhanced digital presence to reach broader audiences
- Improved efficiency in administrative tasks
- Better community engagement tools

Relationship with Community Work:

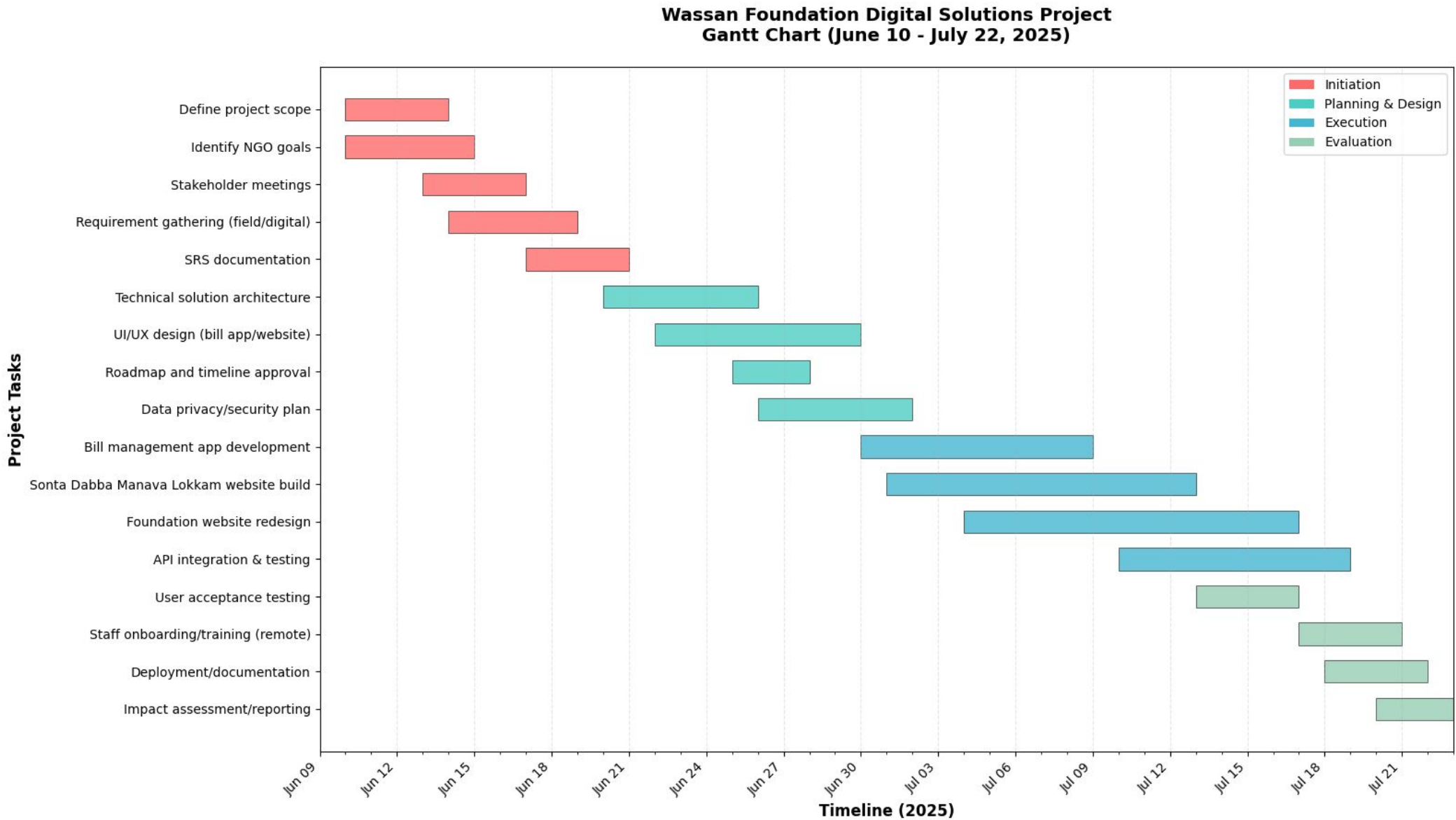
The digital solutions directly support Wassan's mission by improving operational efficiency, allowing more resources to be directed toward community development and natural farming promotion.

Current Stage: Testing and Deployment phase with ongoing impact assessment

Maintenance Plan:

- Training NGO staff on system usage
- Providing technical documentation
- Establishing support protocols
- Regular system updates and improvements

Proposed Milestones and Timelines



Expected Deliverables

Field Engagement

- Engaged in direct community interaction for one day
- Successfully delivered all project milestones through proactive remote collaboration and digital engagement; no on-site field visits required

Leadership & Social Impact

- Led development and delivery of solutions
- Conducted remote training sessions for staff

Technical Deliverables

- Bill management web application
- Sonta Dabba Manava Lokkam website
- Proposed design for wassan foundation website.

Multimedia Documentation

- Github repository of the websites
- Photo documentation
- Certificate



Expected plan to Assess the Impact

Monitoring Strategy:

- Track system usage metrics and efficiency improvements
- Measure reduction in administrative time
- Monitor website engagement and reach

Control Group:

- Compare efficiency before and after digital solution implementation
- Benchmark against similar NGOs without digital solutions

Short-term Impact (1 Year):

- 70% reduction in bill processing time
- Improved financial transparency and accountability
- Enhanced digital outreach reaching 5,000+ individuals
- Direct benefit to 50+ Wassan Foundation staff members

Long-term Impact (3 Years):

- Scalable model adopted by other NGOs in the network
- Increased community engagement in sustainable practices
- Enhanced operational efficiency enabling expanded program reach
- Potential to benefit 25,000+ community members through improved NGO services

Proposed Budget

I.No.	Equipment/Facilities	Qty.	Total (in Rs.)	Justification
1	Development Tools & Software Licenses	1	25,000	Web development frameworks, OCR APIs
2	Cloud Hosting & Domain Services	1 Year	20,000	Website hosting, domain registration
3	Google Workspace Integration	1 Year	15,000	API access, additional storage
4	Testing Devices & Equipment	Set	10,000	Mobile devices, tablets for testing
5	Field Visit & Documentation	-	30,000	Travel, accommodation, documentation
6	Training & Support Materials	-	10,000	User manuals, training resources
		Grand Total	1,10,000	

Concluding Remarks (Further directions)

This project demonstrates the power of technology in amplifying social impact. By developing tailored digital solutions for Wassan Foundation, we've created tools that not only improve operational efficiency but also enhance the organization's ability to promote sustainable agriculture and natural farming practices.

Further Directions:

- Scale the solution to other NGOs in the RRA Network
- Develop mobile applications for field workers
- Integrate AI-powered insights for better decision making
- Create a comprehensive digital ecosystem for the natural farming community

The success of this project lies in its practical approach to solving real-world problems while maintaining alignment with the NGO's core mission of promoting sustainable agricultural practices and community development.